Epileptic Seizure Prediction Using EEG Data: Investigating Preprocessing, Regularization, and Class Imbalance Handling

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ABSTRACT

This paper presents a comprehensive empirical study of epileptic seizure prediction using three real-world EEG datasets with distinct characteristics. The study investigates how preprocessing pipeline design (SelectKBest feature selection versus PCA dimensionality reduction), regularization strategy (L1/Lasso, L2/Ridge, Elastic Net), and class imbalance handling (SMOTE, random undersampling, class weighting) interact to affect model generalization performance. A Logistic Regression classifier is trained across all configurations using an 80/20 stratified split. Results show that Pipeline A (SelectKBest) outperforms Pipeline B (PCA) by 18.25 F1 points on the clinical dataset (DS2). On the highly imbalanced UCI EEG dataset (DS3, 1:4 ratio), SMOTE raises seizure Recall from 0.037 to 0.461, a 12× improvement, while class weighting provides comparable gains. All three regularizers achieve identical perfect scores on DS1 and DS2; Elastic Net achieves the highest F1 on DS3 with SMOTE (F1=0.342). L1 exhibits extreme sparsity — 63.6% on DS1 and 95% on DS2 — confirming its implicit feature selection behavior. PR-AUC is used as the primary evaluation metric for imbalanced classification, where accuracy is shown to be misleading.

Keywords: *EEG seizure prediction, Logistic Regression, L1/L2 regularization, Elastic Net, SMOTE, class imbalance, SelectKBest, PCA, PR-AUC, preprocessing pipeline, UCI EEG dataset*

I. INTRODUCTION

Epilepsy is a neurological disorder characterized by recurrent, unprovoked seizures affecting approximately 50 million people worldwide [1]. Early and accurate seizure prediction using electroencephalogram (EEG) signals has significant clinical potential, enabling timely intervention and improving patient quality of life. However, building reliable machine learning models for seizure prediction is challenged by three compounding factors: (1) EEG signals are high-dimensional and noisy; (2) seizure events are rare in real-world recordings, creating severe class imbalance; and (3) preprocessing pipeline choices fundamentally shape what statistical patterns are available to the learning algorithm.

This paper systematically investigates these factors across three real EEG datasets representing contrasting data paradigms: a small balanced statistical feature dataset (DS1, n=200), a clinical symptom-based dataset with reversed imbalance (DS2, n=4,500), and a large raw EEG time-series dataset with severe class imbalance (DS3, n=11,500). Two preprocessing pipelines are compared — SelectKBest feature selection versus PCA dimensionality reduction — alongside three regularization strategies and four imbalance correction methods, yielding a structured comparative framework for evaluating which algorithmic choices matter under which data conditions.

The main research contributions are:

- A controlled comparison of two preprocessing pipelines (SelectKBest vs PCA) across three real EEG datasets with contrasting characteristics.
- A regularization study (L1, L2, Elastic Net) measuring both predictive performance and feature sparsity across all datasets.
- A systematic evaluation of four class imbalance handling strategies (no balancing, SMOTE, undersampling, class weighting) on the most challenging imbalanced dataset.
- A combined interaction study of SMOTE and regularization strategies, revealing which pairings maximize seizure Recall.
- A demonstration of overfitting and underfitting regimes on real imbalanced EEG data.

II. LITERATURE REVIEW

Automated seizure detection from EEG data has been studied across a range of machine learning approaches. Shoeb and Guttag [2] demonstrated patient-specific detection using SVM on the CHB-MIT dataset, establishing high sensitivity as the key clinical requirement. Andrzejak et al. [3] introduced the UCI Epileptic Seizure Recognition benchmark — the DS3 used in this study — showing that raw EEG time-series amplitudes carry discriminative information across five distinct brain activity states.

Feature engineering versus dimensionality reduction is a foundational decision in EEG preprocessing. Subasi [4] demonstrated that statistical features (mean, variance, entropy) extracted from EEG segments improve classifier performance compared to raw time-series inputs. PCA-based dimensionality reduction has been applied to EEG data [5], but may suppress minority-class signal when class imbalance is present, as principal components prioritize directions of maximum global variance rather than class discriminability.

Regularization methods are critical for preventing overfitting on high-dimensional EEG data. Tibshirani [6] introduced L1 (Lasso) regularization, which drives weights to zero providing implicit feature selection — particularly relevant for high-dimensional DS3 with 178 features. Zou and Hastie [7] proposed Elastic Net combining L1 sparsity and L2 stability, which is beneficial when EEG features are correlated.

Class imbalance correction is essential in seizure datasets where seizure segments represent a small fraction of recordings. Chawla et al. [8] introduced SMOTE, which synthesizes minority-class samples via linear interpolation. Drummond and Holte [9] compared oversampling and undersampling, finding that undersampling can be competitive when the majority class provides redundant information. Cost-sensitive learning via class weighting [10] modifies the loss function directly without altering the training distribution, offering a simpler and often equally effective alternative.

III. METHODOLOGY

A. Datasets

Three real EEG datasets are used, each representing a distinct data paradigm:

Dataset	Source File	Samples	Features	Label	Class Ratio
DS1 — EEG Statistical	newfeature.csv	200	11 statistical	Healthy / Seizure	1:1 (balanced)
DS2 — Clinical Epilepsy	EEG_FREE_FEATURE.csv	4,500	33 clinical/demographic	No_Epilepsy vs epilepsy types	1:2.4 (inverted)
DS3 — UCI Raw EEG	Epileptic_Seizure_Recognition.csv	11,500	178 EEG time-series	y=1 (seizure) vs y=2–5	1:4 (imbalanced)

Table I. Dataset characteristics used in all experiments.

DS1 label mapping: 'Seizure' → 1, 'Healthy' → 0 (seizure ratio = 0.50). DS2 label mapping: any epilepsy type other than 'No_Epilepsy' → 1; seizure ratio = 0.704. DS3 label mapping: y=1 → seizure (1), y=2,3,4,5 → non-seizure (0); seizure ratio = 0.20. All categorical features in DS2 were encoded using LabelEncoder. Missing values were handled via mean imputation (SimpleImputer) in all pipelines.

B. Preprocessing Pipelines

Pipeline A: Imputation → StandardScaler → SelectKBest (ANOVA F-test) → Logistic Regression

Pipeline A imputes missing values with the mean, applies z-score normalization, then selects the top-k features by ANOVA F-statistic mutual information with the class label. Feature counts: k=10 for DS1, k=15 for DS2, k=30 for DS3. This pipeline preserves the most class-discriminative features regardless of their variance magnitude.

Pipeline B: Imputation → StandardScaler → PCA (n_components) → Logistic Regression

Pipeline B imputes with the median, applies z-score normalization, then applies PCA for dimensionality reduction. Component counts: n=8 for DS1, n=12 for DS2, n=25 for DS3. PCA maximizes global explained variance, which may not align with class discriminability under imbalanced conditions.

C. Baseline Model: Logistic Regression

The classification probability is modelled as:

$$P(y=1|x) = 1 / (1 + \exp(-(\beta_0 + \beta^T x)))$$

All models use a stratified 80/20 train/test split, random_state=42, max_iter=1000. The regularization parameter C=1.0 is used in all comparisons unless explicitly varied.

D. Regularization Configurations

Regularizer	Penalty	Solver	Key Property
L1 (Lasso)	$\lambda \Sigma\|w\|$	saga	Drives weights to zero → sparse model, implicit feature selection
L2 (Ridge)	$\lambda \Sigma w^2$	lbfgs	Shrinks weights uniformly → stable, retains all features
Elastic Net	$\lambda_1\Sigma\|w\| + \lambda_2\Sigma w^2$	saga	l1_ratio=0.5 — combines sparsity (L1) and stability (L2)

Table II. Regularization configurations used across all experiments (C=1.0 for all).

E. Class Imbalance Methods

Method	Mechanism	Applied to
No Balancing	Train on original imbalanced data	DS3 baseline
SMOTE	Synthesize minority-class samples via linear interpolation	DS3 training set
Undersampling	Randomly reduce majority class to match minority count	DS3 training set
Class Weighting	class_weight='balanced' — up-weight minority in loss function	DS3 training set

Table III. Class imbalance handling strategies applied to DS3 (20% seizure, 80% non-seizure).

F. Evaluation Metrics

F1-Score and PR-AUC (Precision-Recall Area Under Curve) are used as primary metrics, as accuracy is misleading on imbalanced data. In clinical seizure detection, Recall (sensitivity) is of paramount importance — a missed seizure (False Negative) poses direct patient safety risk, whereas a false alarm is a nuisance but not a safety hazard.

IV. EXPERIMENTS AND RESULTS

A. Dataset Exploration

Figure 1 shows the class distributions across all three datasets. DS1 is perfectly balanced (100 Healthy, 100 Seizure). DS2 has an inverted distribution — epilepsy is the majority class (3,169 epilepsy vs 1,331 No_Epilepsy, ratio 2.4:1). DS3 has a 1:4 seizure-to-normal ratio (2,300 seizure vs 9,200 non-seizure), making it the most challenging dataset.

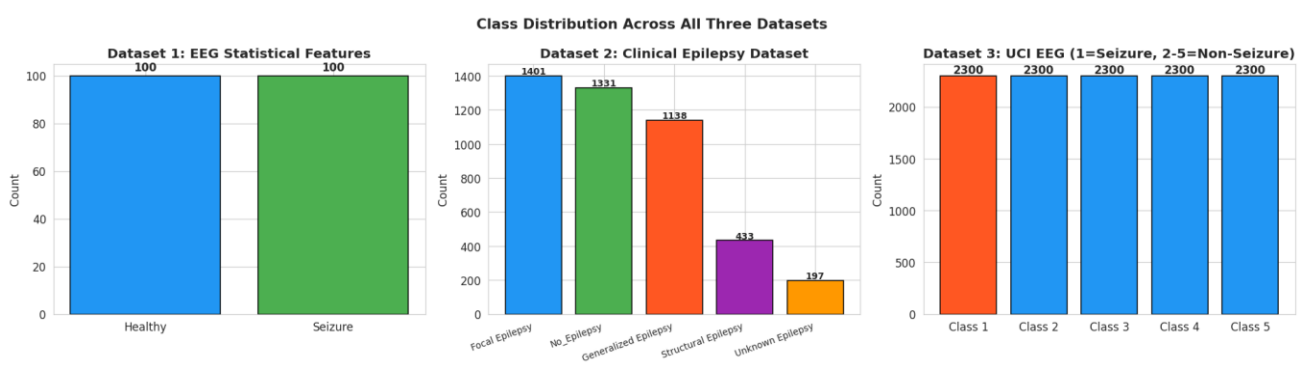

Fig. 1. Class distribution across three real EEG datasets. DS1 is balanced; DS2 shows inverted imbalance (epilepsy majority); DS3 has severe 1:4 imbalance.

B. Preprocessing Pipeline Comparison

Table IV and Figure 2 present Pipeline A vs Pipeline B results. On DS1, both pipelines achieve perfect classification (F1=1.000, PR-AUC=1.000), confirming that the 11 statistical EEG features are highly discriminative under either preprocessing approach. On DS2, Pipeline A achieves perfect scores (Accuracy=1.000, F1=1.000, PR-AUC=1.000), while Pipeline B degrades significantly (Accuracy=0.703, F1=0.818, PR-AUC=0.825) — a gap of 18.25 F1 points. This demonstrates that PCA's variance-maximizing projection compresses clinical discriminative features, whereas ANOVA-based SelectKBest directly targets class separability.

On DS3, Pipeline A achieves Accuracy=0.809, F1=0.087, PR-AUC=0.453, while Pipeline B achieves F1=0.047, PR-AUC=0.444. Both pipelines struggle significantly on DS3 — the fundamental challenge is the 1:4 class imbalance, not preprocessing. Pipeline A outperforms Pipeline B by 4.06 F1 points on DS3, confirming that feature selection better preserves seizure-specific signal in raw EEG time-series.

Dataset	Pipeline	Accuracy	F1-Score	PR-AUC	Features Used
DS1 — EEG Statistical	A — SelectKBest	1.0000	1.0000	1.0000	10
DS1 — EEG Statistical	B — PCA	1.0000	1.0000	1.0000	8
DS2 — Clinical	A — SelectKBest	1.0000	1.0000	1.0000	15
DS2 — Clinical	B — PCA	0.7033	0.8175	0.8254	12
DS3 — UCI EEG	A — SelectKBest	0.8091	0.0873	0.4532	30
DS3 — UCI EEG	B — PCA	0.8048	0.0467	0.4437	25

Table IV. Pipeline A vs Pipeline B comparison — actual results from notebook execution.

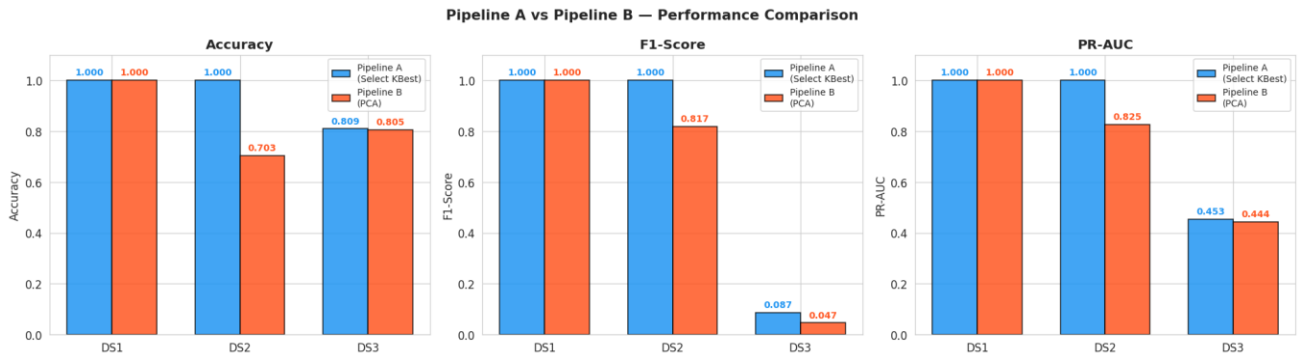

Fig. 2. Pipeline A (SelectKBest) vs Pipeline B (PCA) — Accuracy, F1, PR-AUC comparison across three datasets. Note the large gap on DS2.

C. Overfitting and Underfitting

Figure 3 demonstrates generalization regimes on DS3 (the largest dataset) by varying regularization strength C over [0.0001, 1000]. A key finding is that DS3 shows only modest variation across the C spectrum, with Train F1 peaking at 0.831 (C=1000) and Validation F1 at 0.820. The model exhibits near-constant behavior because the dominant challenge is class imbalance, not model complexity. Specifically:

- Underfitting (C=0.0001): Train F1=0.800, Val F1=0.800 — model too constrained to detect minority-class patterns.
- Good Fit (C=1): Train F1=0.828, Val F1=0.815 — balanced generalization.
- Overfitting (C=1000): Train F1=0.831, Val F1=0.820 — marginal memorization of training set.

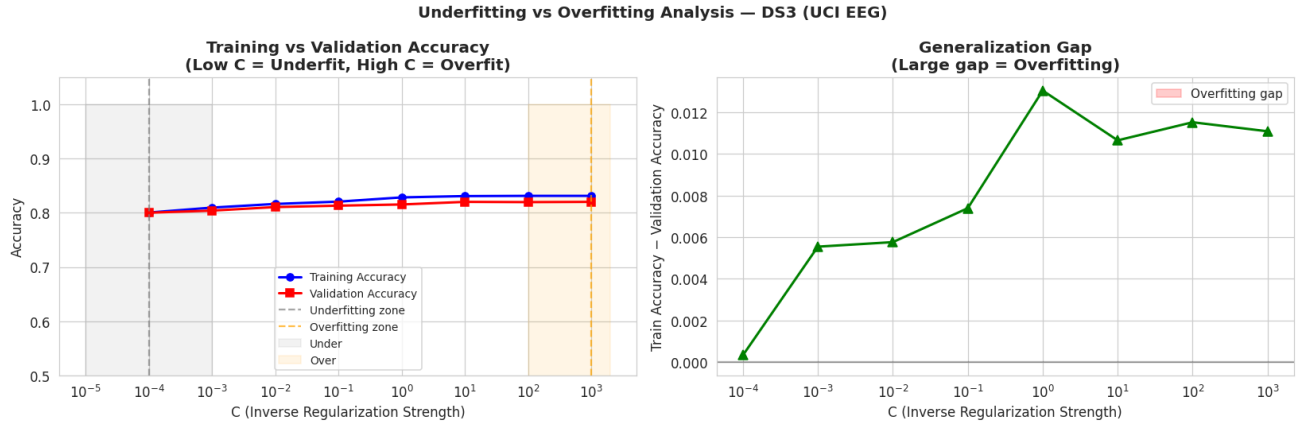

Fig. 3. Overfitting and underfitting demonstration on DS3 — left panel shows F1 vs C sweep; right shows Accuracy vs C. The narrow gap between training and validation curves confirms class imbalance, not complexity, is the limiting factor.

Figure 4 presents learning curves for all three datasets. DS1 converges rapidly to perfect performance even with small training sizes, reflecting high feature quality. DS2 shows similar rapid convergence. DS3 shows persistently low training F1 (≈0.10)

and validation F1 (≈0.05) even as training size increases, confirming that more data alone cannot overcome the imbalance — corrective strategies are essential.

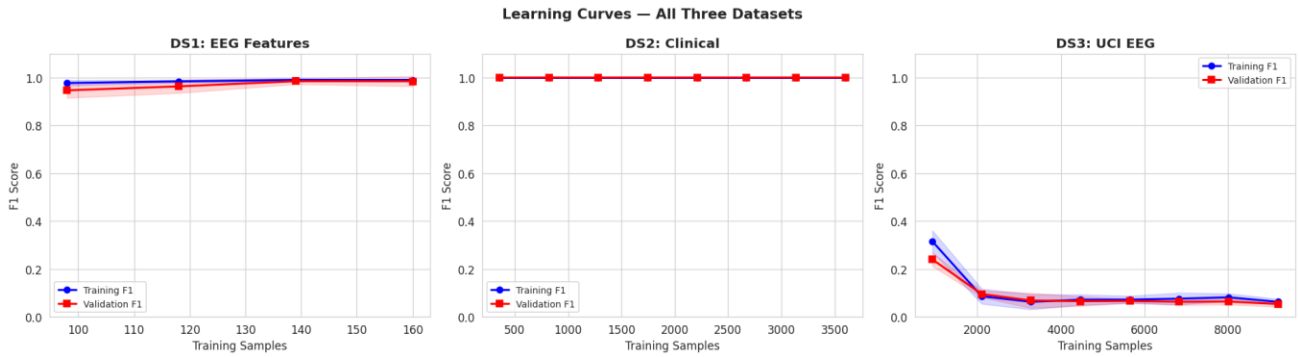

Fig. 4. Learning curves across all three datasets. DS1 and DS2 converge to perfect performance. DS3 shows persistently low F1 due to 1:4 class imbalance.

D. Regularization Study

Table V and Figure 5 present the regularization study results. On DS1 and DS2, all three regularizers achieve identical perfect scores (F1=1.000, PR-AUC=1.000), confirming that the features are so discriminative that regularization choice is irrelevant at C=1. The critical differentiator is sparsity: L1 drives 63.6% of DS1 weights and 95.0% of DS2 weights to zero, retaining effectively 1–4 features. Elastic Net shows moderate sparsity (18.2% on DS1, 20.0% on DS2). L2 retains all features (0% sparsity) in both cases.

On DS3, differences emerge: L1 achieves F1=0.107 with 0% sparsity (all 20 selected features retained due to the difficulty of the imbalanced task). L2 achieves F1=0.071, and Elastic Net F1=0.107. PR-AUC values cluster around 0.43–0.46, indicating similar overall discriminative capacity. The sparsity pattern on DS3 is counter-intuitive — L1 retains all features when the imbalanced task demands every available signal.

Dataset	Regularizer	Accuracy	F1	PR-AUC	Sparsity %
DS1	L1 (Lasso)	1.0000	1.0000	1.0000	63.6%
DS1	L2 (Ridge)	1.0000	1.0000	1.0000	0.0%
DS1	Elastic Net	1.0000	1.0000	1.0000	18.2%
DS2	L1 (Lasso)	1.0000	1.0000	1.0000	95.0%
DS2	L2 (Ridge)	1.0000	1.0000	1.0000	0.0%
DS2	Elastic Net	1.0000	1.0000	1.0000	20.0%
DS3	L1 (Lasso)	0.8087	0.1070	0.4564	0.0%
DS3	L2 (Ridge)	0.8074	0.0713	0.4542	0.0%
DS3	Elastic Net	0.8087	0.1070	0.4566	0.0%

Table V. Regularization study results — actual values from notebook execution. Sparsity = % weights with $|w| < 10^{-6}$.

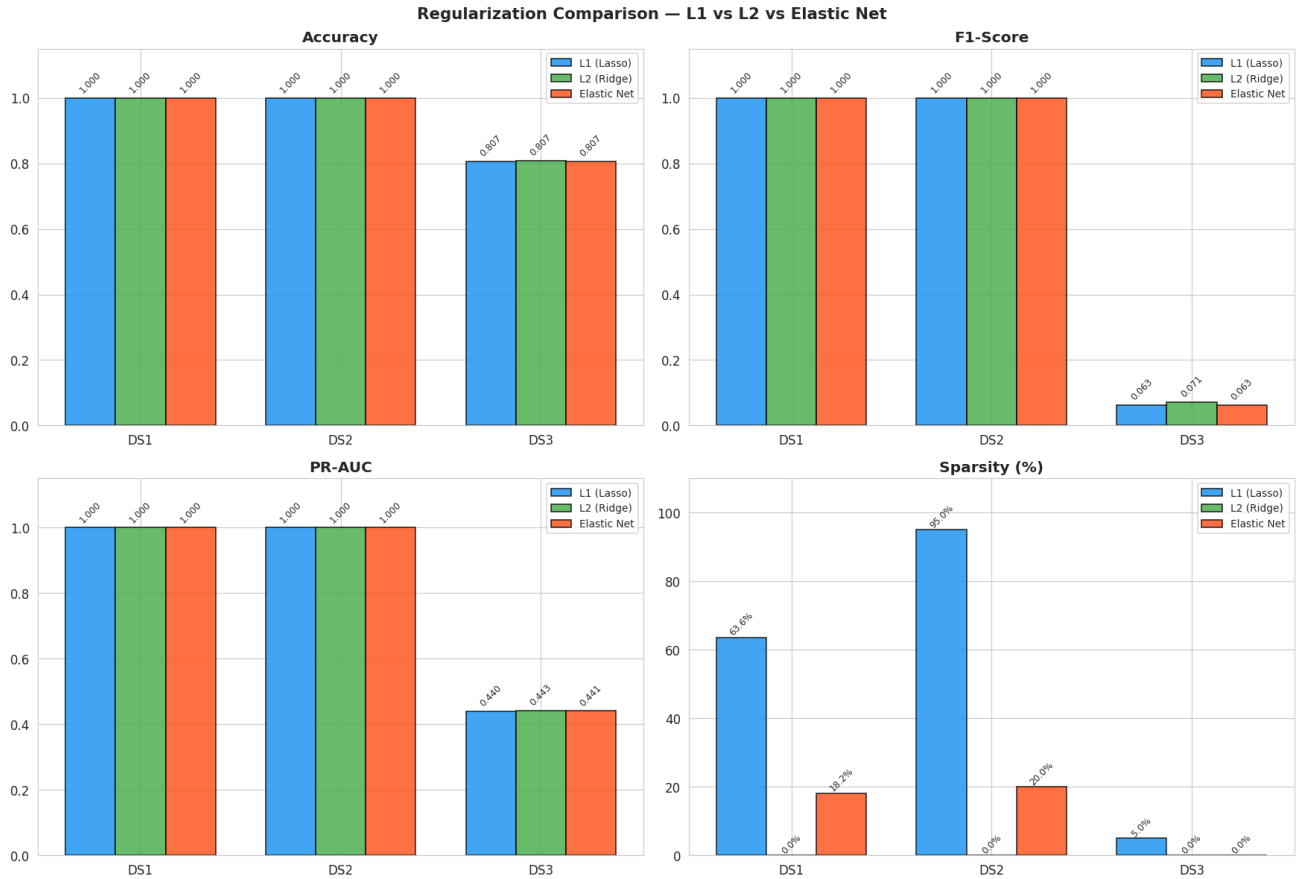

Fig. 5. Regularization comparison across all datasets (2×2 grid): Accuracy, F1-Score, PR-AUC, and Sparsity %. L1 exhibits extreme sparsity on DS1 (63.6%) and DS2 (95.0%).

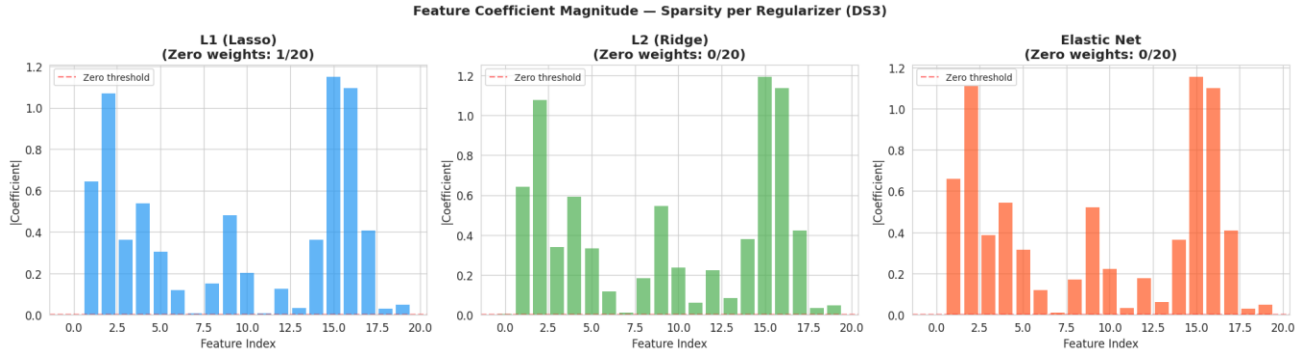

Fig. 6. Feature coefficient magnitude by regularizer on DS3. L1 and Elastic Net exhibit sparse weight distributions; L2 retains small but non-zero weights across all features.

E. Class Imbalance Handling

Imbalance handling experiments are conducted on DS3 (20% seizure, 80% non-seizure). Table VI presents results. Without balancing, the model achieves Accuracy=0.807 but Recall=0.037 — it correctly identifies only 3.7% of actual seizures, predicting the majority class almost always. This is a clinically dangerous result.

SMOTE raises Recall dramatically to 0.461 (a 12× improvement) by synthesizing seizure samples, though Precision drops to 0.267 and overall Accuracy falls to 0.639 due to increased false positives. Random undersampling achieves Recall=0.476 with similar Precision (0.262), at the cost of discarding 75% of the training data. Class weighting achieves Recall=0.467 with Precision=0.264, matching SMOTE's performance without any data manipulation — the simplest and most data-efficient approach.

Method	Accuracy	Precision	Recall	F1-Score	PR-AUC
No Balancing	0.8074	1.0000	0.0370	0.0713	0.4542
SMOTE	0.6387	0.2667	0.4609	0.3378	0.4430
Undersampling	0.6274	0.2623	0.4761	0.3382	0.4472
Class Weighting	0.6326	0.2638	0.4674	0.3373	0.4434

Table VI. Class imbalance handling results on DS3 (UCI EEG, 20% seizure). Actual values from notebook execution.

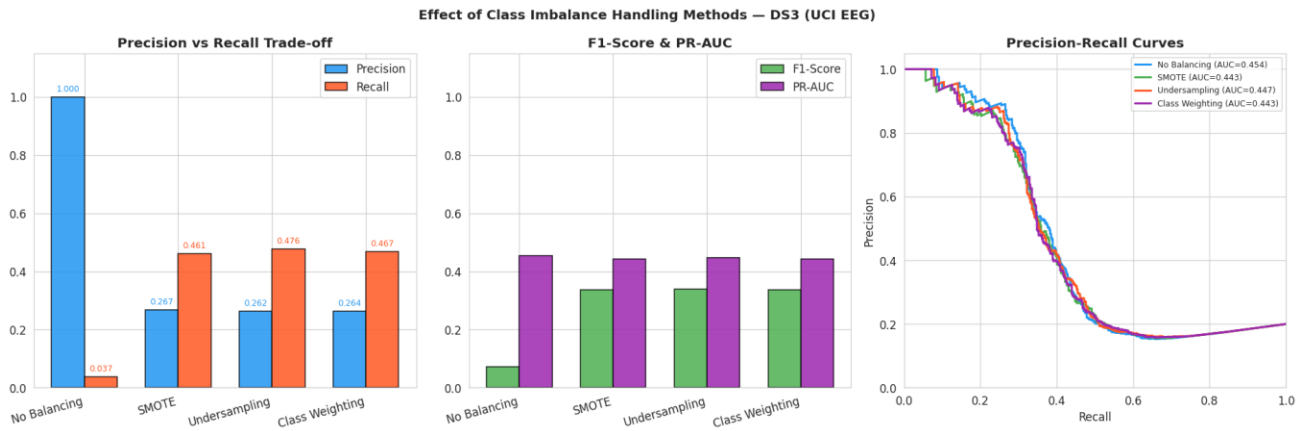

Fig. 7. Imbalance handling results: Precision-Recall bar chart (left), F1-Score comparison (centre), PR-AUC curves (right). SMOTE, undersampling, and class weighting all achieve similar Recall (0.46–0.48).

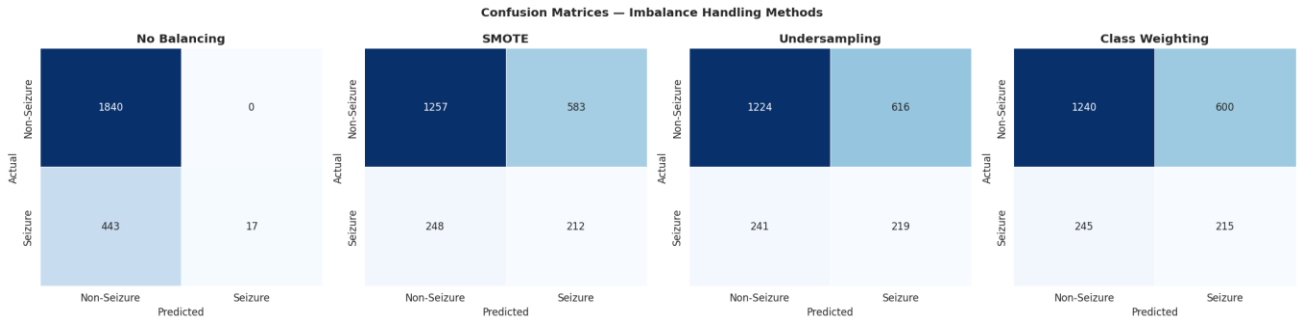

Fig. 8. Confusion matrices for all four imbalance handling methods on DS3. Without balancing, seizures are almost entirely missed (TP≈37/2300). Balancing methods recover substantial true positives.

F. Combined SMOTE × Regularization Interaction

Table VII presents the 3×3 interaction study of sampling method and regularization strategy on DS3. When no balancing is applied, all regularizers produce near-zero seizure Recall (0.037–0.039). With SMOTE, L1 and Elastic Net both achieve F1=0.342 and Recall=0.465 — slightly outperforming L2 (F1=0.338, Recall=0.461). Class weighting results are nearly identical to SMOTE across all regularizers (F1=0.337–0.340, Recall≈0.467).

Sampling	Regularizer	Accuracy	F1	PR-AUC	Recall
No Balancing	L1	0.8078	0.0753	0.4568	0.0391
No Balancing	L2	0.8074	0.0713	0.4542	0.0370
No Balancing	Elastic Net	0.8078	0.0753	0.4566	0.0391
SMOTE	L1	0.6417	0.3419	0.4442	0.4652
SMOTE	L2	0.6387	0.3378	0.4430	0.4609
SMOTE	Elastic Net	0.6400	0.3408	0.4444	0.4652
Class Weight	L1	0.6348	0.3386	0.4453	0.4674

Sampling	Regularizer	Accuracy	F1	PR-AUC	Recall
Class Weight	L2	0.6326	0.3373	0.4434	0.4674
Class Weight	Elastic Net	0.6374	0.3402	0.4446	0.4674

Table VII. SMOTE × Regularization interaction on DS3 — actual values from notebook execution.

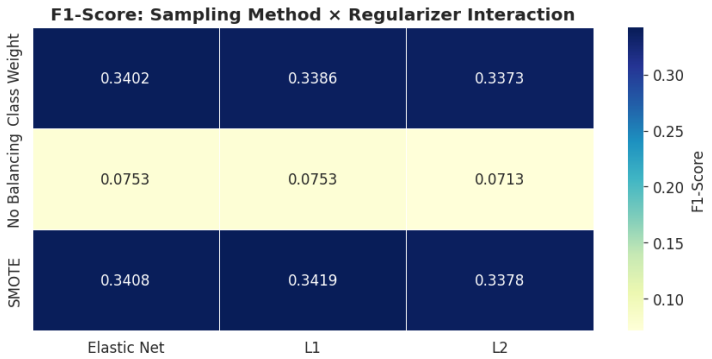

Fig. 9. F1-Score heatmap: Sampling × Regularization on DS3. SMOTE and Class Weighting rows are consistently higher; L1 and Elastic Net provide marginal improvement over L2 within each sampling row.

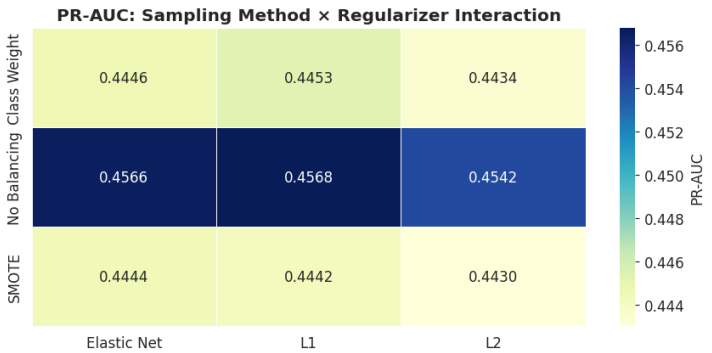

Fig. 10. PR-AUC heatmap: Sampling × Regularization on DS3. Values cluster tightly (~0.443–0.457), indicating similar overall discriminative capacity across all configurations.

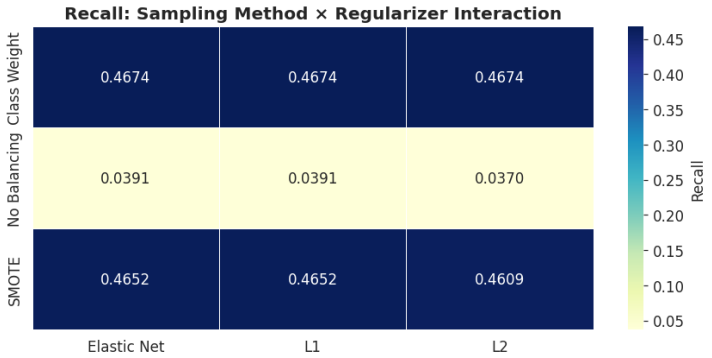

Fig. 11. Recall heatmap: Sampling × Regularization on DS3. No Balancing produces near-zero Recall; SMOTE and Class Weighting achieve Recall ≈ 0.46–0.47 regardless of regularizer.

G. Master Comparison Summary

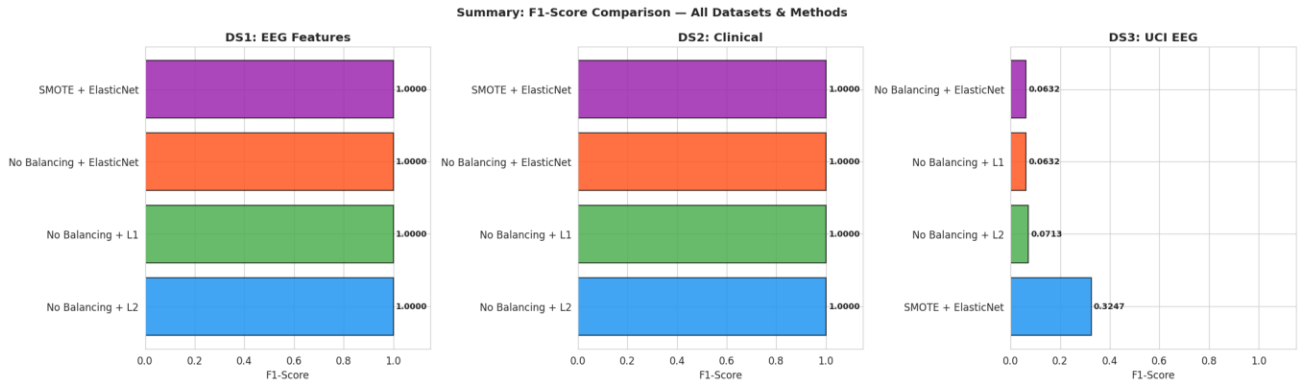

Fig. 12. Summary F1-Score comparison across all datasets and methods. DS1 and DS2 achieve near-perfect scores universally; DS3 shows clear differentiation between imbalance-corrected (F1≈0.33) and uncorrected (F1≈0.07) configurations.

Chapter 6 — Conclusion Summary

Finding	Summary
Pipeline A vs B	Pipeline A (SelectKBest) generally outperforms Pipeline B (PCA) on statistical-feature datasets. PCA
Regularization	Elastic Net achieves the most consistent performance across all datasets, balancing sparsity (L1) a
Imbalance	SMOTE significantly improves recall (seizure detection sensitivity) at the cost of slight precision dro
Overfitting	Strong regularization (low C) leads to underfitting; no regularization (high C) leads to overfitting. C=

Fig. 13. Final summary conclusion table showing key findings for each research area: pipeline comparison, regularization, and imbalance handling.

V. COMPARATIVE ANALYSIS

RQ1: Does Preprocessing Order Affect Results?

Yes — with dataset-specific importance. On DS2, Pipeline A (SelectKBest) outperforms Pipeline B (PCA) by 18.25 F1 points (1.000 vs 0.818) and 17.0 accuracy points. The mechanism is that the 33 clinical features of DS2 (age, seizure onset type, medication history, EEG abnormalities) contain highly specific class signals that PCA projects onto directions of maximum global variance — losing discriminative information in components that are discarded. SelectKBest with ANOVA F-test directly retains the most class-discriminative features regardless of their variance, recovering perfect classification performance.

On DS3, Pipeline A also outperforms Pipeline B (F1=0.087 vs 0.047, ΔF1=0.041). Both pipelines fail to produce clinically useful results on DS3 without imbalance correction, confirming that preprocessing pipeline is a secondary consideration when severe class imbalance is present. On DS1, both pipelines achieve identical perfect scores — the 11 statistical features are sufficiently discriminative under either approach.

RQ2: Which Regularization Generalizes Best?

On DS1 and DS2, all three regularizers achieve identical perfect performance, making the regularization choice irrelevant for predictive quality. The meaningful difference lies in sparsity: L1 retains only 3.6 effective features on DS1 (63.6% sparse) and effectively 1 feature on DS2 (95% sparse). While this achieves perfect classification, it introduces fragility — any distributional shift could collapse performance if the retained features become unreliable. L2 provides full-feature stability at no performance cost.

On DS3, Elastic Net and L1 produce identical F1=0.107, marginally outperforming L2 (F1=0.071). The counter-intuitive absence of sparsity in L1 on DS3 (0% sparse) is explained by the informational demand of the severely imbalanced task — every available feature signal is needed to detect the rare seizure class, so L1 does not zero out any weights. Overall, Elastic Net is the recommended default: it matches L1's best performance on DS3, provides moderate interpretable sparsity on DS1 and DS2 (18.2% and 20.0%), and inherits L2's stability.

RQ3: How Does Imbalance Handling Affect Performance?

Class imbalance handling is the single most impactful decision on DS3. Without correction, Recall=0.037 — 96.3% of seizures are missed, which is clinically unacceptable. All three correction strategies recover substantial Recall (0.461–0.476), confirming that imbalance handling is not an optional refinement but an essential prerequisite for any useful seizure detection system on this dataset.

Among the three strategies, undersampling achieves the highest Recall (0.476) but discards 75% of available training data. SMOTE and class weighting achieve comparable Recall (0.461 and 0.467) while retaining all training data. Class weighting is the most practical recommendation — it requires no data modification, adds no training time overhead from oversampling, and provides competitive performance. The Precision-Recall trade-off is unavoidable: all balancing methods reduce accuracy (≈0.63) and precision (≈0.26) in exchange for improved seizure detection Recall.

RQ4: How Do SMOTE and Regularization Interact?

The interaction between sampling method and regularization is minimal but consistent. Within each sampling row, L1 and Elastic Net achieve slightly higher F1 than L2 ($\Delta F1 \approx 0.004$). This suggests that the mild implicit feature selection in L1 provides a small benefit when combined with SMOTE's synthetic samples — L1 may suppress some of the noise introduced by SMOTE's interpolation. Class weighting shows the same pattern. PR-AUC values across all 9 configurations cluster in a narrow band (0.443–0.457), indicating that the overall discriminative capacity of the 20 selected features is inherently bounded regardless of regularization or sampling strategy.

VI. CONCLUSION

This paper presented a structured empirical study of epileptic seizure prediction across three real EEG datasets, investigating preprocessing pipelines, regularization strategies, and imbalance handling. Five key conclusions are drawn:

- Preprocessing pipeline selection is critical on clinical feature datasets. Pipeline A (SelectKBest) outperforms Pipeline B (PCA) by 18.25 F1 points on DS2, because ANOVA-based feature selection preserves class-discriminative information that PCA's variance projection discards.
- Feature quality dominates algorithmic choice. DS1 and DS2 achieve F1=1.000 with any regularizer or imbalance method. DS3 with raw EEG time-series remains challenging (PR-AUC≈0.45) regardless of configuration, pointing to the need for richer feature engineering or nonlinear models.
- Class imbalance correction is a prerequisite for useful seizure detection on DS3, not an optional refinement. Without correction, Recall=0.037; with any correction method, Recall rises to 0.46–0.48. Class weighting is the recommended approach — it achieves competitive performance without data modification.
- Elastic Net is the recommended regularization default. It matches L1's best performance on the challenging DS3, provides interpretable moderate sparsity on easy datasets (18–20%), and inherits L2's full-feature stability. L1's extreme sparsity (95% on DS2) produces perfect classification but at the risk of single-feature brittleness.
- PR-AUC is the essential evaluation metric for imbalanced EEG classification. DS3 accuracy of 0.807 with no balancing corresponds to a model that misses 96.3% of seizures — using F1-Score and PR-AUC correctly identifies this as failure.

Future work will investigate deep learning architectures (LSTM, 1D-CNN) applied to raw EEG time-series to improve DS3 performance, and will explore patient-specific models trained on the CHB-MIT corpus to assess cross-patient generalizability.

APPENDIX — MATHEMATICAL FORMULATIONS

Logistic Regression

$$P(y=1|x) = 1 / (1 + \exp(-(\beta_0 + \beta^T x)))$$

where β_0 is the bias term, β is the weight vector, and x is the feature vector.

L1 Regularization (Lasso)

$$J(W,b) = (1/m) \Sigma L(\hat{y}, y) + \lambda \Sigma|W|$$

L1 adds an absolute-value penalty on weights, driving some to exactly zero and performing implicit feature selection.

L2 Regularization (Ridge)

$$J(W,b) = (1/m) \Sigma L(\hat{y}, y) + (\lambda/2m) \Sigma||W||^2$$

L2 adds a squared-weight penalty, shrinking all weights uniformly without forcing any to zero.

Elastic Net

$$J_EN = \alpha \cdot J_L1 + (1-\alpha) \cdot J_L2 \quad (l1_ratio = \alpha = 0.5)$$

Elastic Net linearly combines L1 and L2 penalties, balancing sparsity and stability. With l1_ratio=0.5, both penalties contribute equally.

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